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B.Tech. Degree I Semester Examination November 2019

CS/EC/IT 19-200-0102 B ENGINEERING PHYSICS

(2019 Scheme)

Time : 3 Hours

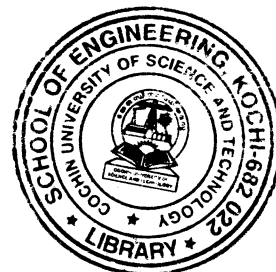
Maximum Marks : 60

PART A

(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Explain the recording of the hologram.
- (b) What is the basic principle of fibre optics?
- (c) What are the important space lattices in the cubic system? Give their characteristics?
- (d) Mention any three properties of liquid crystals.
- (e) Write a note on fullerene.
- (f) Explain the concept of penetration depth in superconductivity.
- (g) State and explain Uncertainty Principle in quantum mechanics.
- (h) The intensity of sound produced by thunder is 0.1 W/m^2 . Calculate the intensity level in decibels.



PART B

(4 × 12 = 48)

- II. (a) Explain with a neat diagram, the construction and working of He-Ne laser. (8)
- (b) What fraction of sodium atom is in the first excited state in a sodium vapour lamp at a temperature of 250°C ? (4)

OR

- III. (a) Define numerical aperture and derive an expression for numerical aperture and acceptance angle of fibre. (8)
- (b) A step index fibre has $n_1 = 1.68$, $n_2 = 1.44$. Calculate critical angle, maximum angle of refraction, acceptance angle and numerical aperture. (4)
- IV. (a) What are Miller Indices? How will you determine the Miller indices of a given plane? (8)
- (b) In a simple cubic crystal, find the ratio of intercepts on the three axes by (123) plane and also find the ratio of spacing of (110) and (111) planes. (4)

OR

- V. (a) Write an essay about the properties of metallic glasses and their applications. (8)
- (b) What is shape memory effect? (4)
- VI. (a) Explain any four properties of nanomaterials in detail. (8)
- (b) What is quantum confinement effect in nanomaterials? (4)

OR

- VII. (a) Describe BCS theory briefly? (8)
- (b) For a specimen, the critical fields are $1.4 \times 10^5 \text{ A/m}$ and $4.2 \times 10^5 \text{ A/m}$ for 14.2 K and 13.7 K. Calculate the critical temperature and critical magnetic field at 0 K and 4.2 K. (4)
- VIII. (a) Derive Schrodinger's time dependent wave equation. (8)
- (b) Calculate the velocity and kinetic energy of an electron of wavelength $1.66 \times 10^{-10} \text{ m}$. (4)

OR

- IX. (a) What is magnetostriction effect? Describe the production of ultrasonic waves by magnetostriction method. (8)
- (b) A quartz crystal of thickness 2mm is vibrating at resonance. Calculate the fundamental frequency and first overtone. Given Young's modulus for quartz $7.9 \times 10^{10} \text{ N/m}^2$ and density for quartz 2650 kg/m^3 . (4)